

WILLISTON SLOULIN FIELD, ND, USA

WMO: 727670

Lat: 48.174N

Lon: 103.637W

Elev: 580

StdP: 94.55

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94014

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.6	(c) -26.7	(d) -32.9	(e) 0.2	(f) -29.1	(g) -29.9	(h) 0.3	(i) -26.3	(j) 13.9	(k) -7.6	(l) 12.0	(m) -6.0	(n) 2.2	(o) 200	(p) 0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	34.3	19.6	32.2	19.2	30.2	18.6	21.9	29.8	20.8	28.8	19.8	27.5	4.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	15.2	25.4	18.1	14.0	24.6	17.0	13.0	23.5	66.9	29.9	62.8	28.7	59.0	27.5	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	10.6	9.0	-33.0	38.1	3.1	1.6	-35.3	39.3	-37.1	40.2	-38.9	41.1	-41.2	42.3
			-33.1	24.2	3.0	1.4	-35.2	25.2	-37.0	26.0	-38.6	26.8	-40.8	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-11.2	-9.1	-2.0	5.9	12.0	17.6	21.7	20.7	14.8	6.6	-2.0	-8.8
	DBStd	12.96	8.47	7.94	7.80	5.56	4.50	3.89	3.38	3.69	4.78	5.22	6.66	7.59
	HDD10.0	2823	656	536	373	142	30	2	0	0	12	128	361	583
	HDD18.3	4911	914	769	629	372	201	58	8	18	125	363	611	841
	CDD10.0	1218	0	0	2	20	93	228	362	333	156	24	1	0
	CDD18.3	264	0	0	0	0	5	35	112	92	19	1	0	0
	CDH23.3	3614	0	0	1	21	123	494	1388	1210	352	25	0	0
	CDH26.7	1480	0	0	0	4	29	159	608	541	134	4	0	0
(14) Wind	WSAvg	3.9	3.7	3.7	4.1	4.5	4.5	4.1	3.7	3.6	3.7	3.9	3.7	3.6
(15) Precipitation	PrecAvg	366	14	11	17	30	53	64	57	36	31	22	14	15
	PrecMax	555	45	44	57	84	188	157	168	118	90	90	84	50
	PrecMin	232	1	1	0	1	4	11	6	1	1	0	0	1
	PrecStd	83	10	10	12	21	36	33	39	25	23	19	13	9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	9.3	20.1	26.4	30.0	33.8	37.1	37.1	34.1	26.5	17.1	8.5
		MCWB	3.8	4.4	10.1	13.3	16.4	20.1	20.5	19.7	18.5	14.1	9.3	3.9
	2%	DB	4.1	5.8	15.2	21.8	26.7	30.6	34.4	34.3	30.4	22.0	12.9	5.1
		MCWB	1.5	2.6	7.5	11.2	15.2	18.8	20.2	19.3	17.1	12.5	6.6	2.0
	5%	DB	2.5	3.5	11.4	18.9	24.1	28.2	32.2	32.0	27.5	18.4	9.8	3.0
		MCWB	0.6	1.2	5.8	9.7	13.9	17.9	19.8	18.7	16.1	10.4	4.8	0.7
	10%	DB	0.6	1.4	8.1	16.2	21.5	26.2	30.2	29.7	24.5	15.4	6.8	1.2
		MCWB	-0.9	-0.1	4.1	8.3	12.7	16.8	19.4	18.2	15.0	9.1	3.2	-0.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	4.9	10.3	13.7	18.1	22.7	24.1	22.8	20.1	15.5	9.7	4.1
		MCDB	7.0	8.5	19.1	23.8	25.7	30.2	31.2	30.9	29.1	24.1	16.4	7.6
	2%	WB	1.8	2.8	8.0	11.8	16.4	20.6	22.4	21.1	18.5	13.0	7.1	2.3
		MCDB	3.8	5.6	14.5	20.4	24.2	27.5	30.3	30.3	27.4	20.4	12.3	4.8
	5%	WB	0.6	1.4	6.0	10.3	14.9	19.1	21.3	20.1	17.1	11.2	5.1	0.9
		MCDB	2.4	3.2	11.0	17.9	21.9	26.0	29.0	28.7	25.3	17.1	9.3	2.9
	10%	WB	-0.8	0.0	4.2	8.8	13.6	17.8	20.3	19.1	15.7	9.6	3.4	-0.6
		MCDB	0.6	1.5	7.8	15.4	20.2	24.6	27.8	27.3	23.1	14.9	6.5	1.2
(35) Mean Daily Temperature Range	MDBR		10.6	10.7	11.5	13.4	13.8	13.4	14.6	15.5	15.0	12.8	11.1	10.2
	5% DB	MCDBR	11.3	12.0	16.8	18.6	18.3	17.1	17.6	18.8	20.5	18.8	15.8	11.9
		MCWBR	9.1	9.1	10.1	9.9	9.0	7.8	6.8	7.2	9.2	10.2	10.2	9.2
	5% WB	MCDBR	11.0	11.4	16.0	17.3	15.5	14.6	15.0	16.0	17.5	17.4	15.1	11.7
		MCWBR	9.1	8.7	9.9	9.6	8.1	7.5	6.7	6.9	8.2	9.6	10.0	9.2
(40) Clear-Sky Solar Irradiance	taub		0.253	0.264	0.278	0.326	0.341	0.356	0.365	0.374	0.328	0.294	0.262	0.248
	taud		2.445	2.462	2.527	2.405	2.423	2.423	2.419	2.368	2.465	2.549	2.519	2.473
	Ebn at Noon		846	915	954	927	921	905	891	866	881	867	831	809
	Edh at Noon		62	77	86	108	110	111	110	111	91	71	58	54
(44) All-Sky Solar Radiation	RadAvg		1.53	2.56	3.82	4.88	5.82	6.37	6.80	5.56	4.12	2.56	1.55	1.11
	RadStd		0.15	0.26	0.31	0.28	0.44	0.45	0.30	0.36	0.35	0.27	0.20	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page